

Yoel Kim

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Research Interests

- Software Engineering → Formal software verification; Software reverse engineering; Embedded software.
- Programming Languages → Static and dynamic program analyses; Programming-by-example.
- Theory of Computation → Automata learning; Abstraction.

Education

- Ph.D. in Computer Science and Engineering, Kyungpook National University* *Mar 2023 – Feb 2027
(Expected)*
- Advisor: Yunja Choi
 - Dissertation: Automated Behavioral Model Inference from Stateful C Programs
- M.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2021 – Feb 2023*
- Advisor: Yunja Choi
 - Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification
- B.S. in Computer Science and Engineering, Kyungpook National University* *Mar 2017 – Feb 2021*
- Track: Global Software Convergence
 - GPA: 3.95/4.3, Rank: 1/13

Ongoing Work

1. AUTOCL: A Tool for Active Learning of Symbolic Automata from C Programs
Yoel Kim and Yunja Choi
(In preparation) **TACAS'27: International Conference on Tools and Algorithms for the Construction and Analysis of Systems** (Regular Tool Papers Track)
2. Learning Functional Specifications from Stateful Control Programs through Active Model Learning
Yunja Choi and Yoel Kim
(Submitted) *IEEE Access*, 2026

International Conferences

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper
Yoel Kim and Yunja Choi
FSE'26: ACM International Conference on the Foundations of Software Engineering
2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software
Yoel Kim and Yunja Choi
FSE'24: ACM International Conference on the Foundations of Software Engineering

Domestic Conferences

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
Seungbin Choi, Yoel Kim, and Yunja Choi
KCSE'26: *Korea Conference on Software Engineering* * *Short Paper Award*
2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation
Yoel Kim and Yunja Choi
KCSE'23: *Korea Conference on Software Engineering* * *Best Short Paper Award*
3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis
Yoel Kim and Yunja Choi
KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain
Yoel Kim and Yunja Choi
KCSE'21: *Korea Conference on Software Engineering*

Domestic Journals

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
(Extended version of the KCSE'26 paper)
Seungbin Choi, Yoel Kim, and Yunja Choi
KIPS Transactions on Software and Data Engineering, 2026

Talks

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Top Conference Session. KNU CSE BK21 Graduate Student Exchange Conference. *Daegu, Korea. Aug 12, 2026.*
2. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Research Paper Presentation. FSE'26. *Concordia University, Montreal, Canada. Jul 7, 2026.*
3. AUTOCL: Active Learning of Symbolic Automata via Dynamic Symbolic Mapper. Milestone Talk. 10th STAAR Workshop. *Ulsan, Korea. Feb 3, 2026.*
4. Active Learning of Symbolic Automata via Dynamic Symbolic Mapper. PhD Presentation Session. KNU CSE BK21 Graduate Student Exchange Conference. *Daegu, Korea. Jan 15, 2026.*
5. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. Milestone Talk. 9th STAAR Workshop. *Korea University, Seoul, Korea. Jul 29, 2025.*
6. C2FSM: A Tool for Active Learning of Extended Finite State Machines from C Programs. PhD Presentation Session. KNU CSE BK21 Graduate Student Exchange Conference. *Daegu, Korea. Jul 21, 2025.*
7. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCC'25. *Jeju, Korea. Jul 4, 2025.*
8. Efficient Active Model Learning using Program Synthesis. Lightning Talk. 8th STAAR Workshop. *Yeosu, Korea. Feb 7, 2025.*
9. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*
10. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KNU CSE BK21 Graduate Student Exchange Conference. *Daegu, Korea. Jan 21, 2025.*
11. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research Paper Presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*
12. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Published Paper Session. 7th STAAR Workshop. *Gyeongju, Korea. Jul 9, 2024.*
13. Automatic Generation of Software Disaster Models via Active Learning. Lightning Talk. 6th STAAR Workshop. *KAIST, Daejeon, Korea. Jan 30, 2024.*
14. PBE-Based Function Summarization and Refinement for Efficient Property Falsification. Lightning Talk. 5th STAAR Workshop. *Ulsan, Korea. Jul 5, 2023.*
15. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation. Research Paper Presentation. KCSE'23. *Pyeongchang, Korea. Feb 10, 2023.*
16. PBE-Based Automatic Stub Generation and Stub Code Partitioning for Efficient Model Checking. Lightning Talk. 4th STAAR Workshop. *Paju, Korea. Feb 1, 2023.*
17. Improving Model Checking Efficiency Using Program Synthesis and Counterexample Trace-Based Alarm Filtering. Lightning Talk. 3rd STAAR Workshop. *Pohang, Korea. Jul 14, 2022.*

18. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis. Research Paper Presentation. KSC'21. *Pyeongchang, Korea. Dec 21, 2021.*
19. A Case Study on the Performance of Program Synthesis in Embedded Software Domain. Research Paper Presentation. KCSE'21. *Online. Feb 2, 2021.*

Experience

Teaching Assistant , Kyungpook National University	<i>Mar 2021 – Jun 2024</i>
◦ ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).	
◦ COMP0224: Software Design (Fall 2021, 2022).	
◦ COMP0216: Data Structure Applications (Spring 2021).	
Undergraduate Tutor , Kyungpook National University	<i>Mar 2018 – Dec 2020</i>
◦ COMP0322: Database Management Systems (Fall 2020).	
◦ CLTR266: Software Computational Thinking (Spring 2018, 2019).	
◦ CLTR268: Python Programming (Winter 2018).	
◦ COMP0216: Data Structure Applications (Fall 2018).	
Undergraduate Research Intern , Software Safety Engineering Lab	<i>Jul 2019 – Dec 2019; Sep 2020 – Feb 2021</i>
◦ Supervisor: Yunja Choi	
◦ Developed a Java-based GUI tool to support software testing activities for Hyundai Motor.	
Intern , JS System, Daegu	<i>Dec 2019 – Feb 2020</i>
◦ Assisted with testing an FPGA-based embedded board.	
Undergraduate Research Intern , AI-Based Networks Lab	<i>Oct 2018 – Dec 2018</i>
◦ Supervisor: Dongkyun Kim	